CSCE 581: Introduction to Trusted AI

Lectures 15 and 16: Mitigations, LLMs, Project Update #2

PROF. BIPLAV SRIVASTAVA, AI INSTITUTE

3RD AND 5TH MAR, 2026

Carolinian Creed: "I will practice personal and academic integrity."

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Organization of Lectures 15, 16

- Introduction Section
  - Recap from Week 7 (Lectures 13 and 14)
  - Announcements and News
- Main Section
  - L15: Explanation, Rating, LLMs / FMs
  - L16: Project Update #2
- Concluding Section
  - About next week/ non-holiday – Lectures 17, 18
  - Ask me anything

Introduction Section

Recap from Week 7 (Lectures 13, 14)

- We looked at
  - Explanations – LIME method
  - Transparency through documentation – Rating – ARC tool

Announcement: Guest Speaker

Individual and Collective Human Agency in the Face of AI, by Dr. Kush Varshney

March 17, 2026

Dr. Varshney is an IBM Fellow based at the Thomas J. Watson Research Center, Yorktown Heights, NY, where he directs Human-Centered Trustworthy AI research. He applies data science and predictive analytics to human capital management, healthcare, olfaction, computational creativity, public affairs, international development, and algorithmic fairness, which has led to the Extraordinary IBM Research Technical Accomplishment for contributions to workforce innovation and enterprise transformation, and IBM Corporate Technical Awards for Trustworthy AI and for AI-Powered Employee Journey. He and his team created several well-known open-source toolkits, including AI Fairness 360, AI Explainability 360, Uncertainty Quantification 360, and AI FactSheets 360. AI Fairness 360 has been recognized by the Harvard Kennedy School's Belfer Center as a tech spotlight runner-up and by the Falling Walls Science Symposium as a winning science and innovation management breakthrough.

He independently-published a book entitled 'Trustworthy Machine Learning' in 2022, available at http://www.trustworthymachinelearning.com. He is a fellow of the IEEE.
Personal website: https://krvarshney.github.io

Discussion Quiz and Project

- Quizzes
  - Quiz-1 Assessment
  - Quiz 2
- Project Update #2

#1 NEWS – Prompt Repetition Improves Non-Reasoning LLMs

Link - https://arxiv.org/pdf/2512.14982v1

- **Context**
  - LLMS are given prompts to get outputs
  - " <QUERY> " to " <QUERY><QUERY> ".
  - LLMs are mostly trained as causal models

- **Findings**
  - "<CONTEXT> <QUESTION>" often performs differently from a query of the form "<QUESTION> <CONTEXT>"
  - Changed " <QUERY> " to " <QUERY><QUERY>
  - Tried options-first vs. question-first
- **Insight**
  - For non-reasoning, repetition helps; three and other variations also help. ***Prompt repetition wins 47 out of 70 tests, with 0 losses.***
  - For reasoning, repetition does not help, or is limited. These models usually repeat questions anyway and that could be a reason for no significant impact

#2 NEWS – Anthropic Safety and Usage in US Govt/ Defense

Link - https://www.americanprogress.org/article/the-trump-administration-is-trying-to-make-an-example-of-the-ai-giant-anthropic/ ; https://www.anthropic.com/news/building-safeguards-for-claude

- **Context**
  - Anthropic claims to build safeguards in Claude output; high performer in many tasks
  - Safeguards rarely independently audited; evolving environment and regulations
  - US Govt using/ wants to use Claude; contentious use – on US internal matters, autonomous killing

- **Findings**
  - Govt wants Anthropic to drop safeguards
  - Little technical merit or justification given by US govt
  - Anthropic refused to drop safeguards – not clear for all or just for US govt
  - US signed up OpenAI instead; 6 months to phase out/ phase In

- **Insight**
  - Claiming to have or dropping safeguards does not make an impact; performance does
  - **All comes down to 'trust'**

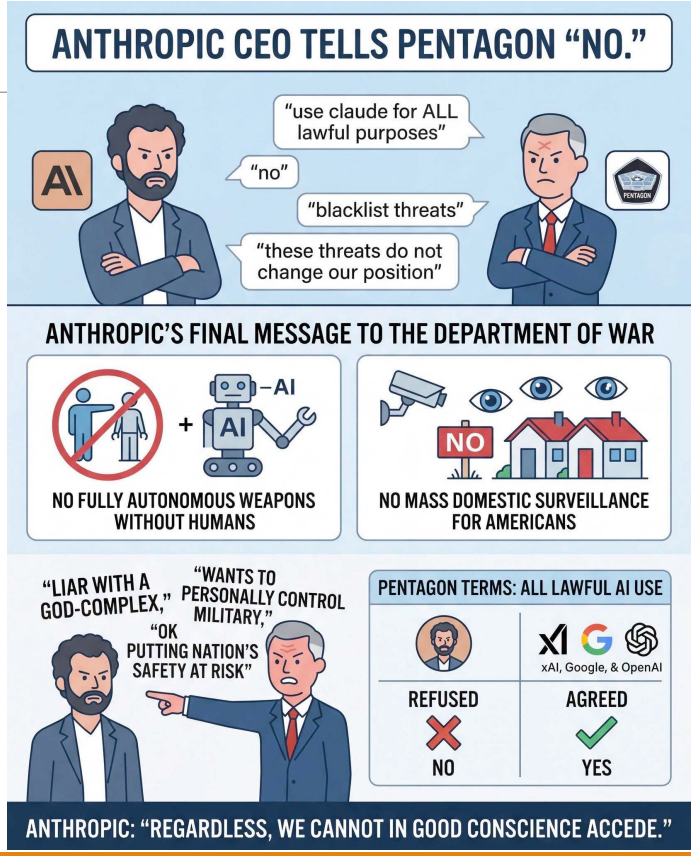


Image Credit: from FB

Valuable Reading

- AAAI 2025 Presidential Panel on the Future of AI: 17 topics related to AI research, each chapter sketching its history, current trends and open challenges. Has insights both from experts and survey respondents. https://aaai.org/wp-content/uploads/2025/03/AAAI-2025-PresPanel-Report-FINAL.pdf

Update:
Administrative Information – CSCE 581

Walk through of Github: https://github.com/biplav-s/course-tai-s26/

Website: https://sites.google.com/site/biplavsrivastava/teaching/ai-csce-581-spring-2026-trusted-ai

Quick Info - When and Where

•Tuesday/Thursday 11:40 am – 12:55 pm

•In person at **Swearingen Engr Ctr| Room 2A15**. Recordings will be available on Blackboard whenever feasible.

Catalog Information

• Trusted AI - CSCE 581 001 CRN: 53067

•Duration: 01/12/2026 - 05/06/2026

Instructor Information

•Instructor: Biplav Srivastava, E-mail: biplav.s AT sc.edu

•Office Hours: 10:15-11:15am (Tu, Th); other times by appointment

•**GitHub for slides, sample code**.

TA Information

• Bharath Muppasani, bharath@email.sc.edu; Github ID – **BharathMuppasani**

•Office Hours: 11-12 (M), 2-3pm (W); Storey Building, Room 1209

Student Assessment

A = [920-1000]
B+ = [870-919]
B = [820-869]
C+ = [770-819]
C = [720-769]
D+ = [670-719]
D = [600-669]
F = [0-599]

Tests	Undergrad	Grad
Course Project – report, in-class presentation	600	600
Quiz – best of 2 from 3	200	200
Final Exam	200	100
Additional Final Exam – Paper summary, in-class presentation		100
Total	1000 points	1000 points

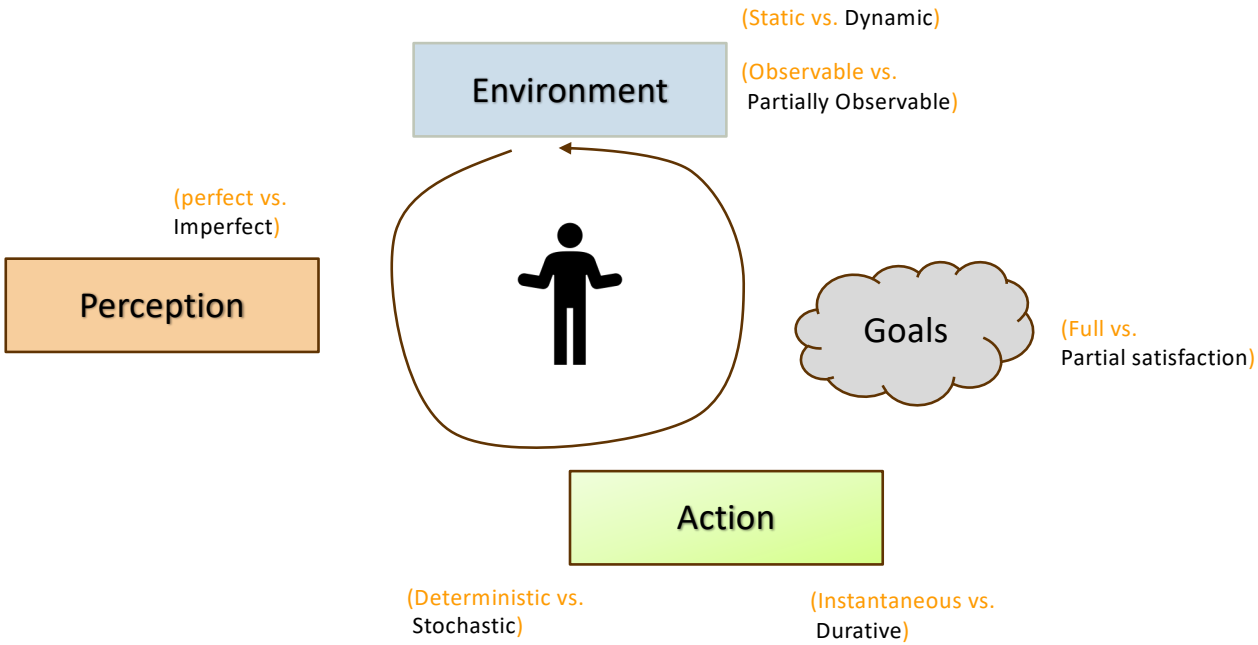
Grades and Attendance Policy

As an in-person class, I will impose a **common-sense**, **necessary – but not sufficient**, attendance policy for those taking the class for grades.

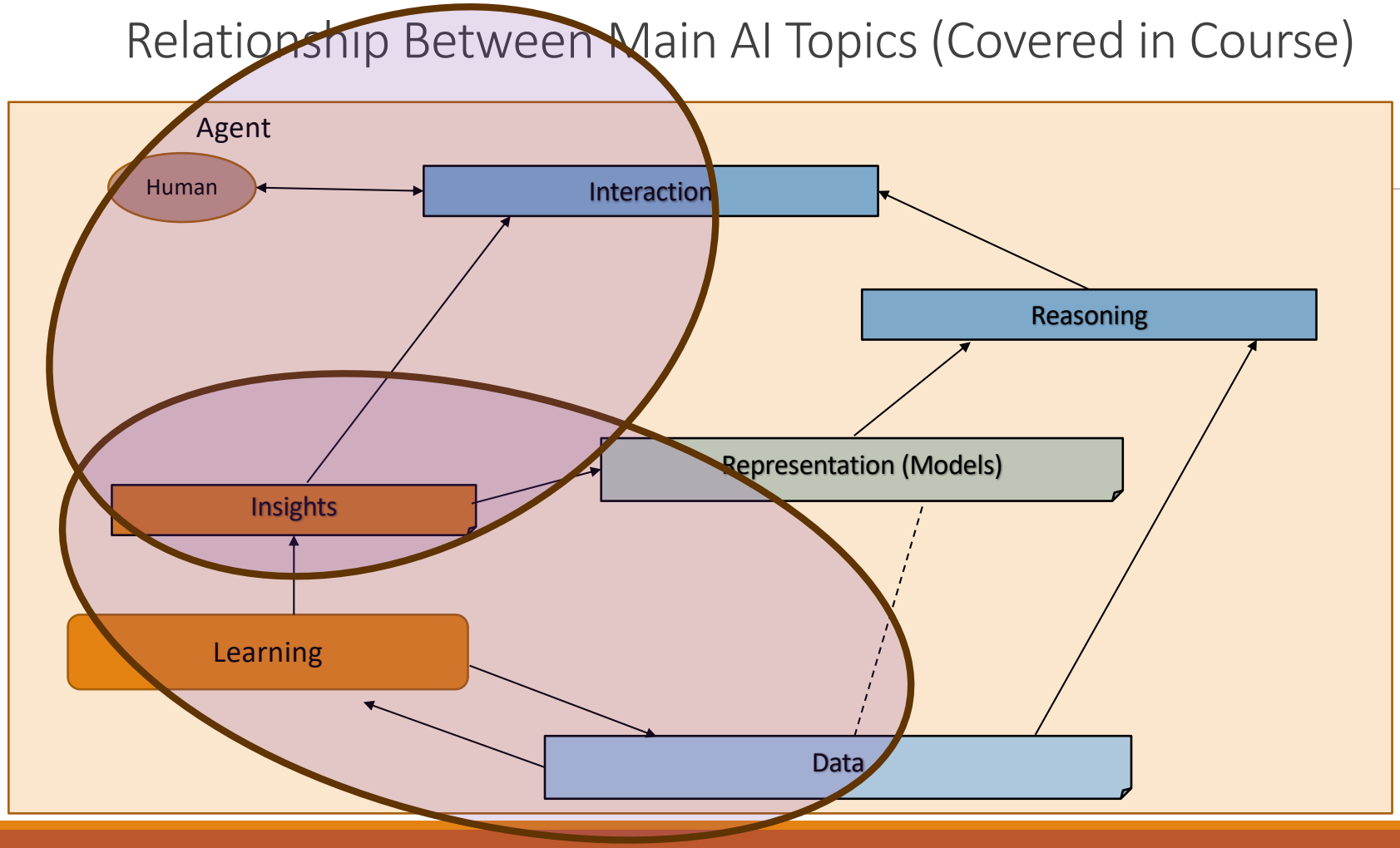
will be out of 14 classes:

1. 80% or more classes attending: **A, B+ or B**
2. 50-79% classes: **B+, B, C+, C, D+ or D**
3. 49% or below: **F**

Intelligent Agent Model



Relationship Between Main AI Topics (Covered in Course)



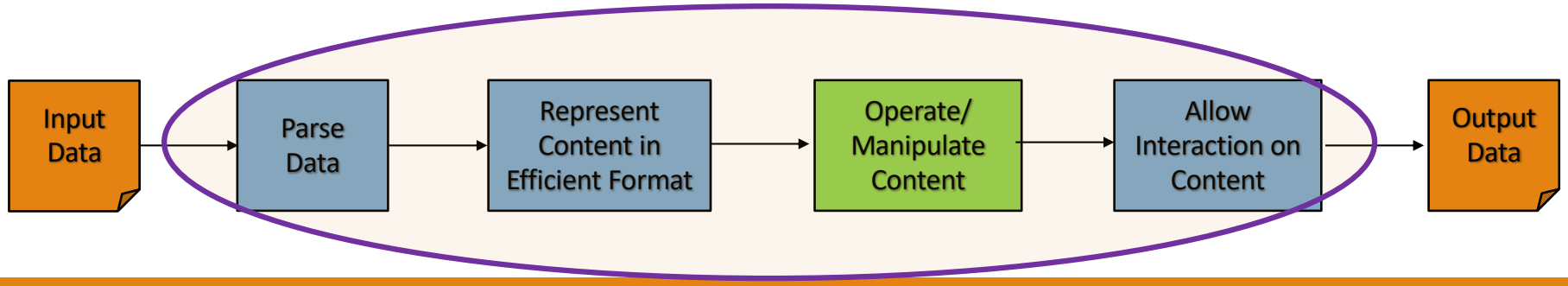
High Level Semester Plan (Adapted, Approximate)

CSCE 581 –

- Week 1: Introduction
- Week 2: Background: AI - Common Methods
- Week 3: The Trust Problem
- Week 4: Machine Learning (Structured data) - Classification
- Week 5: Machine Learning (Structured data) - Classification – Trust Issues
- Week 6: Machine Learning (Structured data) – Classification – Mitigation Methods
- Week 7: Machine Learning (Structured data) – Classification – Explanation Methods
- Week 8: Machine Learning (Text data, **vision**) – Classification, **Large Language Models**
- Week 9: Machine Learning (Text data) - Classification – Trust Issues, LLMs
- Week 10: Machine Learning (Text data) – Classification – Mitigation Methods
- Week 11: Machine Learning (Text data) – Classification – Explanation Methods
- Week 12: Emerging Standards and Laws, **Real world applications**
- Week 13: Project presentations
- Week 14: Project presentations, Conclusion

AI/ ML topics and with a focus on fairness, explanation, Data privacy, reliability

Main Segment



InterpretML

- **Details**: https://github.com/interpretml/interpret
  - Whitebox (Glassbox) models: change learning code to introduce explainability support
  - Blackbox models: don't change learning code

Interpretability Technique	Type
Explainable Boosting	**glassbox model**
APLR	glassbox model
Decision Tree	glassbox model
Decision Rule List	glassbox model
Linear/Logistic Regression	glassbox model
SHAP Kernel Explainer	blackbox explainer
LIME	**blackbox explainer**
Morris Sensitivity Analysis	blackbox explainer
Partial Dependence	blackbox explainer

InterpretML – Sample Code

Github: https://github.com/biplav-s/course-ai-f24/blob/main/sample-code/l21-explainability/ExploreInterpreatbilityPackageMS.ipynb

Explanation Taxonomy

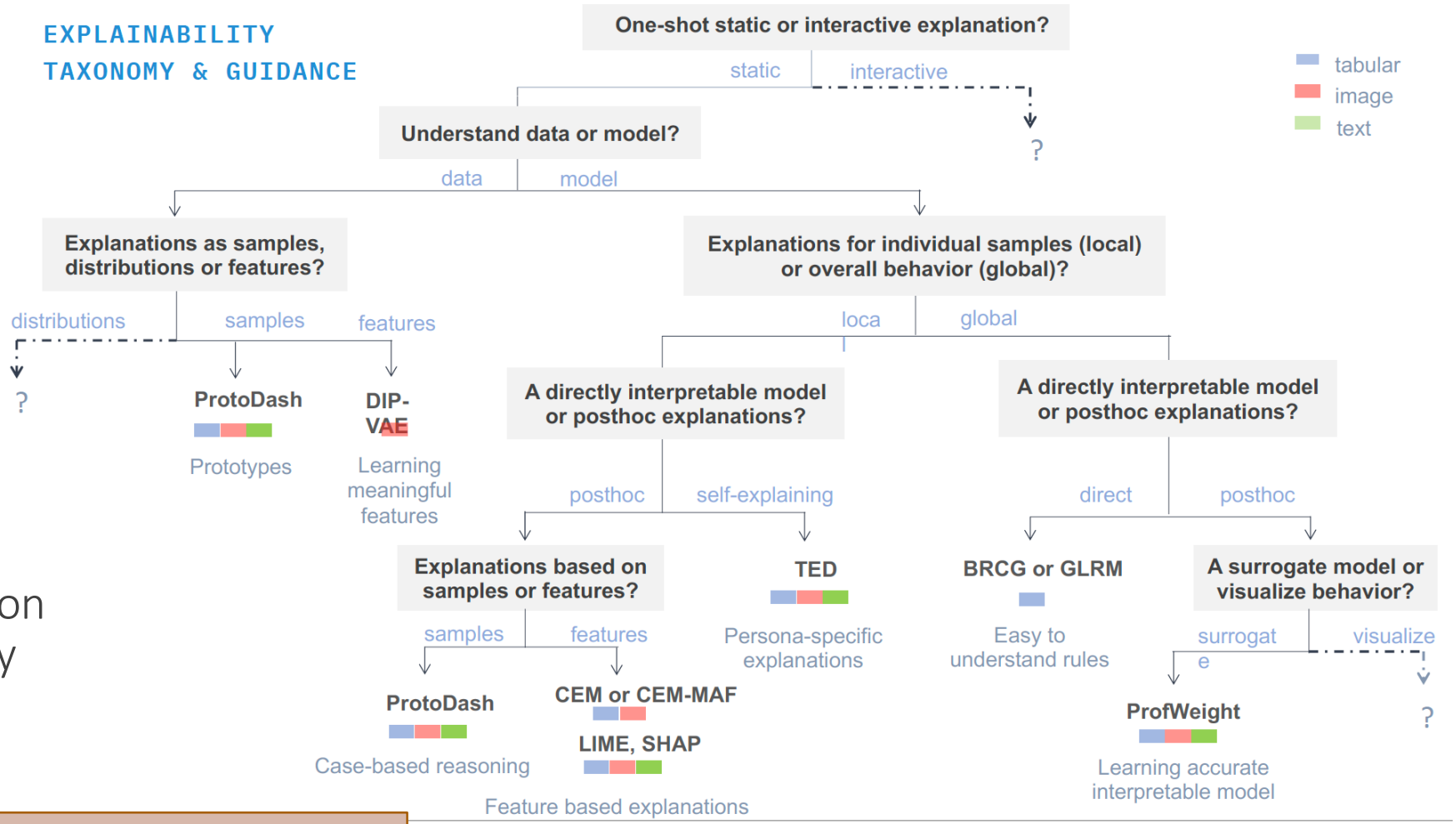
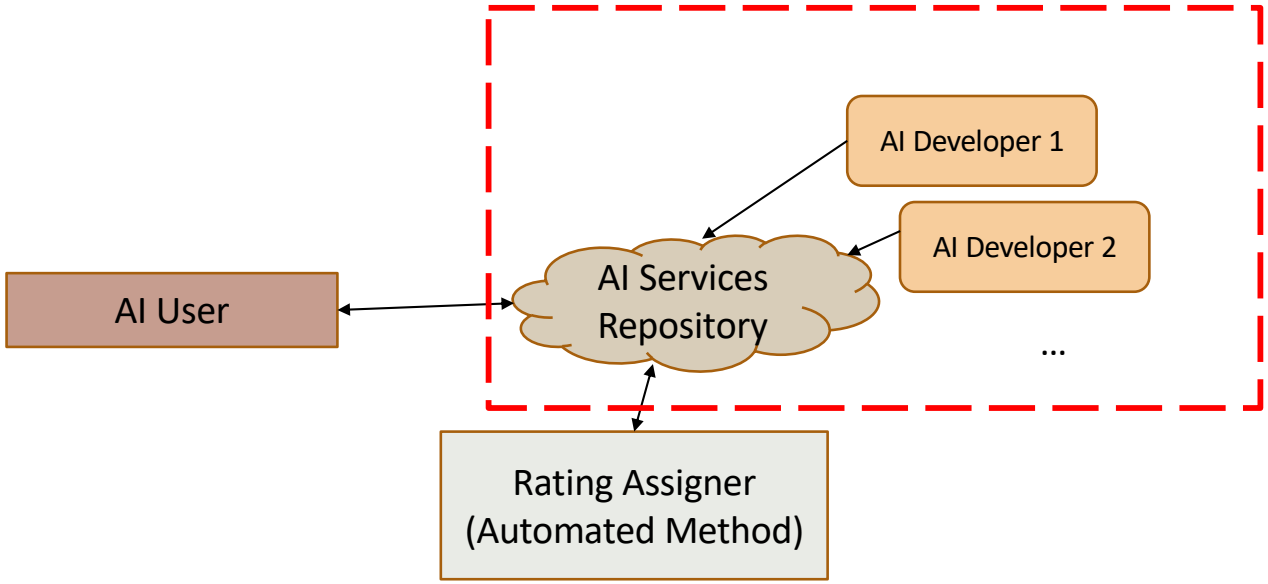


Figure Credit: Diptikalyan Saha and Vijay Arya, Oct 2021

Rating (Certification) of Models

Generating Ratings / Certificates for AI's Assessed Behavior

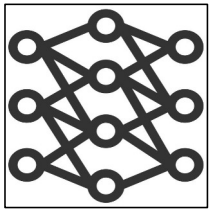
Idea: Develop Automated Methods to Rate AI Systems That Can be Used for Communicating Trust in Black Box Setting



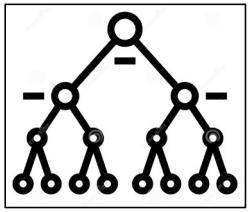
Black-Box and White-Box AI Models

- **Black-box models** often produce accurate predictions but **do not reveal their decision logic**, making them harder to interpret.
- **White-box models** allow us to inspect how each input contributes to the final decision and they are **easier to interpret**.
- **Many recent AI models fall into the first category**. These models require separate methods to explain their behavior (Ex: eXplainable AI (XAI) methods, causal models, …).

Examples:



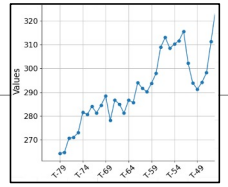
Black-Box: Deep Neural Networks (DNNs)

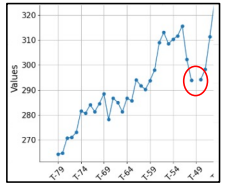


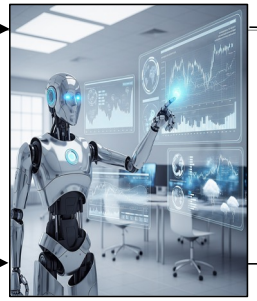
White-Box: Decision Trees

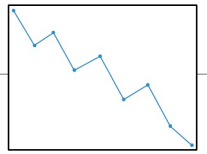
Trust Me, I am AI (But Should You?)

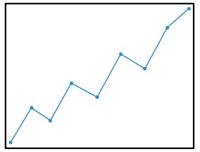
Time-Series Forecaster









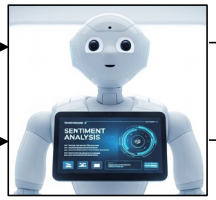


One missing value can throw off the entire prediction!

Sentiment Analyzer

Amanda is feeling **depressed**.

Adam is feeling **depressed**.



Sentiment: 0

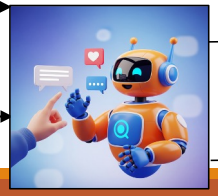
Sentiment: -0.4

Protected information affecting the predictions!

Chatbot

My name is **Alonzo**. **What is the capital of South Carolina?**

My name is **Jack**. **What is the capital of South Carolina?**



The capital of South Carolina is Columbia. SC became the first state to ratify the Articles of Confederation ...

The capital of South Carolina is Columbia.

Trust Issues – Mitigate via Ratings

- Communicate behavior via certificates / ratings (increase transparency)
  - But, let humans make decisions

Trust Dimensions
Competent
Reliable
Upholds human values
Allows human interaction

Transparency Through Documentation of Rating

Documentation about

- Outcome (e.g., Nutrition label, Electronic DataSheet, Factsheet)
- Process (e.g., SEI Capability Maturity Model, ISO 9001)

Documentation by

- Producer (e.g., Nutrition label)
- Consumer (e.g., Yelp rating)
- Independent 3rd Party (e.g., JD Powers, NHTSA car crash)

Reference: AboutML Project at PAI - https://www.partnershiponai.org/about-ml-get-involved/#read

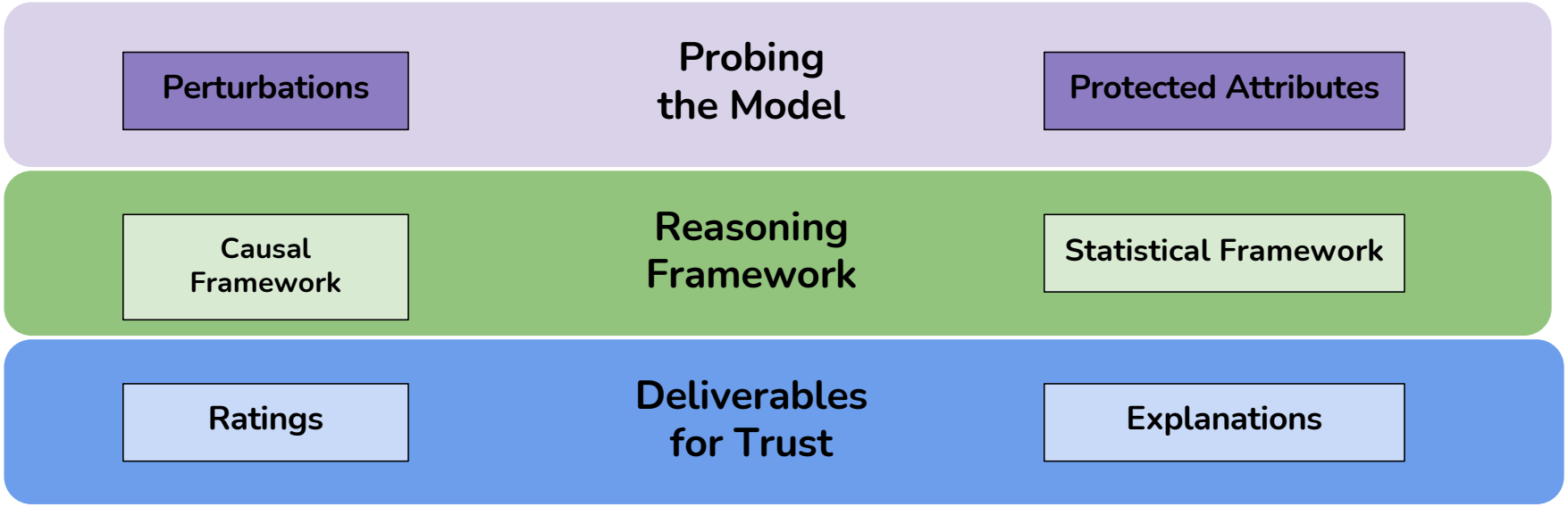
Explanation v/s Rating (Certification)

- Explanation
  - Focused on model behavior with respect to given input data
  - May give local or global explanation

- Certification
  - Gives general expected behavior
  - Global in nature; not specific to a particular data / input

- Complementary roles

Understanding and Comparing AI Model Behavior



Problem We Are Tackling for AI

Insight

- *Empower people to make informed decisions regarding which AI to choose*
- *Communicate trust information better!*
  - *Analogy: Food labels*
- *Facilitate users in understanding their choices*

Calories 230	Calories from Fat 40
	% Daily Value*
Total Fat 8g	12%
Saturated Fat 1g	5%
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 160mg	7%
Total Carbohydrate 37g	12%
Dietary Fiber 4g	16%
Sugars 1g	
Protein 3g	
Vitamin A	10%
Vitamin C	8%
Calcium	20%
Iron	45%

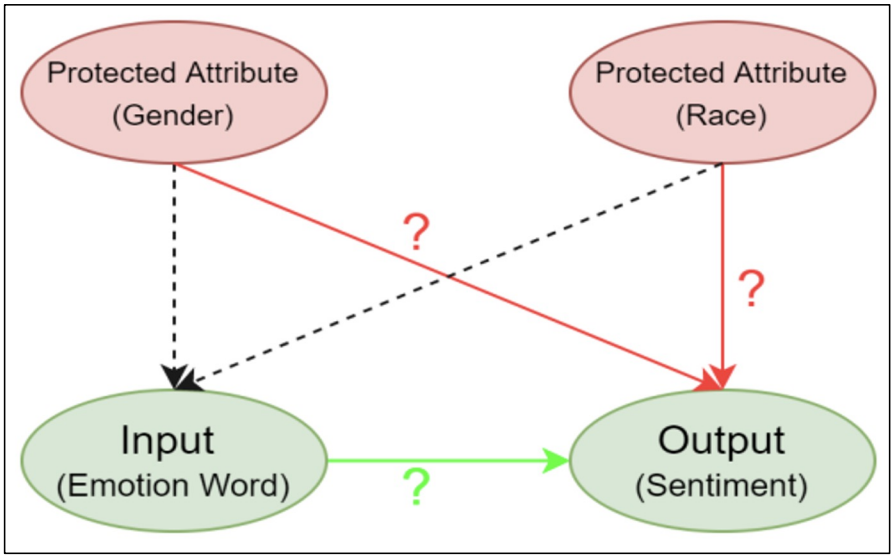
* Percent Daily Values are based on a 2,000 calorie diet. Your daily value may be higher or lower depending on

Amount per 2/3 cup	Calories 230
% DV*	
12%	**Total Fat** 8g
5%	Saturated Fat 1g
	Trans Fat 0g
0%	**Cholesterol** 0mg
7%	**Sodium** 160mg
12%	**Total Carbs** 37g
14%	Dietary Fiber 4g
	Sugars 1g
	Added Sugars 0g
	Protein 3g
10%	**Vitamin D** 2mcg
20%	**Calcium** 260mg

In a series of previous work, we have developed ideas for ***rating bias of AI services and robustness***

: created new methods and supported evaluations with further user studies

Key Idea: Causal Setup - Sentiment

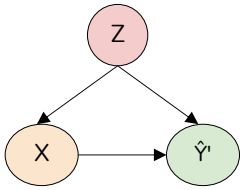


- **Red** and **green** arrows indicate **undesirable** and **desirable** causal paths, respectively.
- The '?' indicates causal links that require verification, which our rating approach assists with.

Rating AI Models: Choose Your AI Model Like You Choose Your Peanut Butter!

Ratings | Deliverables for Trust | Explanations

Input: Test data and models.
Hypothesis: *Attribute Z* influences the models' *Predictions* $\hat{Y}$.



Rating Method

Input: Test data and models.
Hypothesis: *Attribute X* influences the models' *Predictions* $\hat{Y}$.

Output: Rating is given to different binary classifiers.

Model	Raw Scores	Rating
Logistic (Interpretable)	0	1
Random (Non-Interpretable)	1.6	2
Random Forest (Non-Interpretable)	3.9	3
Biased (Interpretable)	4.6	4

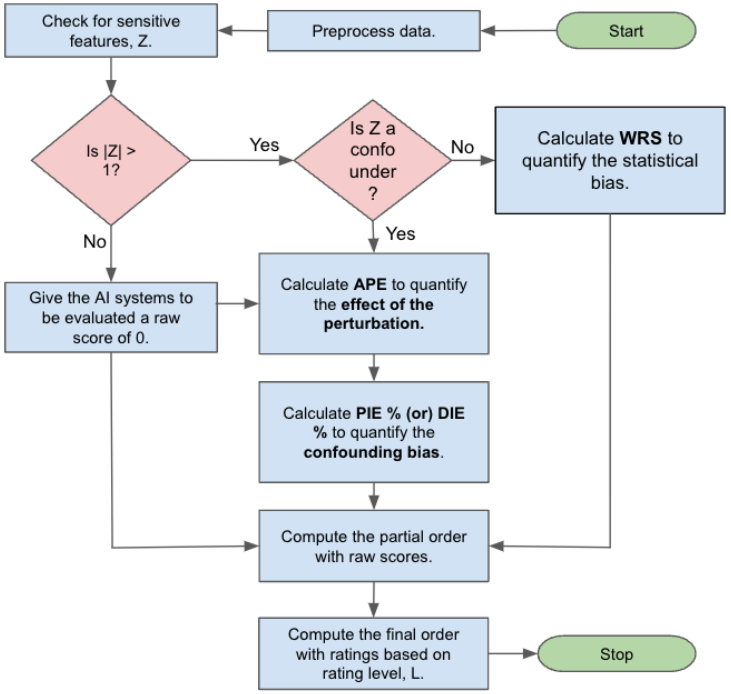
Output: Rating is given to different binary classifiers.

Model	Raw Scores	Rating
Random Forest (Non-Interpretable)	0.3	1
Logistic (Interpretable)	0.4	2
Biased (Interpretable)	0.6	3
Random (Non-Interpretable)	0.7	4

ARC Tool and Rating

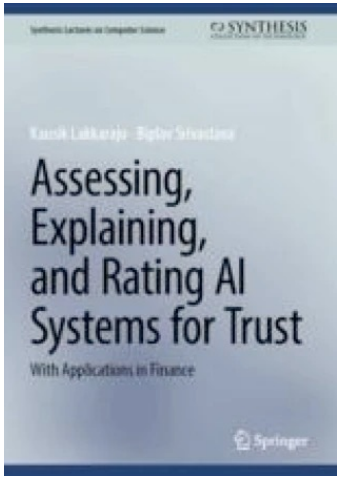
Tool: http://casy.cse.sc.edu/causal_rating

- See Sentiment Assessment Systems (SAS)



More Details

- **Assessing, Explaining, and Rating AI Systems for Trust,** With Applications in Finance, Kausik Lakkaraju , Biplav Srivastava, 2026
  - https://link.springer.com/book/9783032210388
    ISBN: 978-3-032-21038-8



LLMs

Language Model

Problem:
Given a sentence fragment, predict what word(s) come next

Language Model:
estimate probability of substrings of a sentence

$$P(w_i|w_1, w_2, \ldots, w_{i-1}) = \frac{P(w_1, w_2, \ldots, w_{i-1}, w_i)}{P(w_1, w_2, \ldots, w_{i-1})}$$

Bigram approximation

$$P(w_i|w_1, w_2, \ldots, w_{i-1}) \approx \frac{P(w_{i-1}, w_i)}{P(w_{i-1})}$$

From Jurafsky & Martin

Applications:

- Spelling correction
- speech recognition
- machine translation,
- …

Language Model

Markovify library
https://github.com/jsvine/markovify

Language Model:
estimate probability of substrings of a sentence

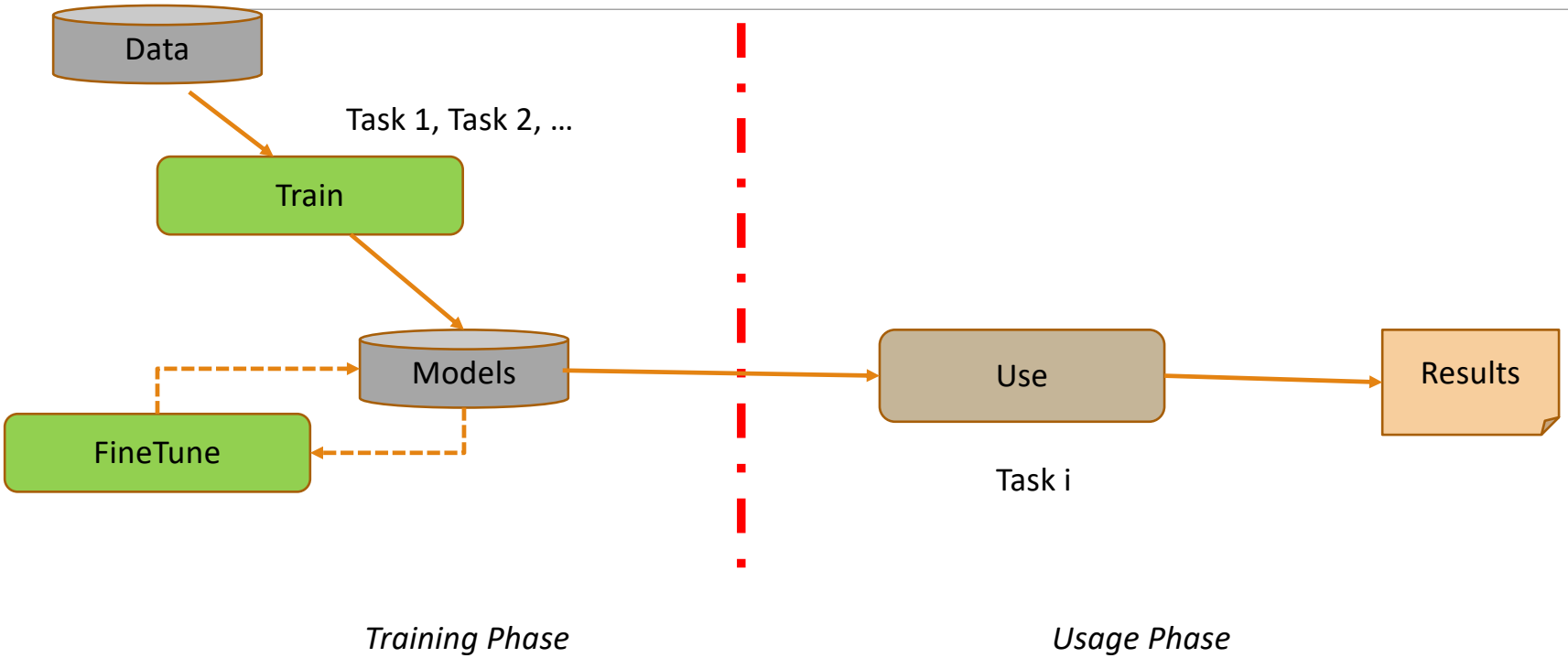
$$P(w_i|w_1, w_2, \ldots, w_{i-1}) = \frac{P(w_1, w_2, \ldots, w_{i-1}, w_i)}{P(w_1, w_2, \ldots, w_{i-1})}$$

See code samples with Markovify library on Github

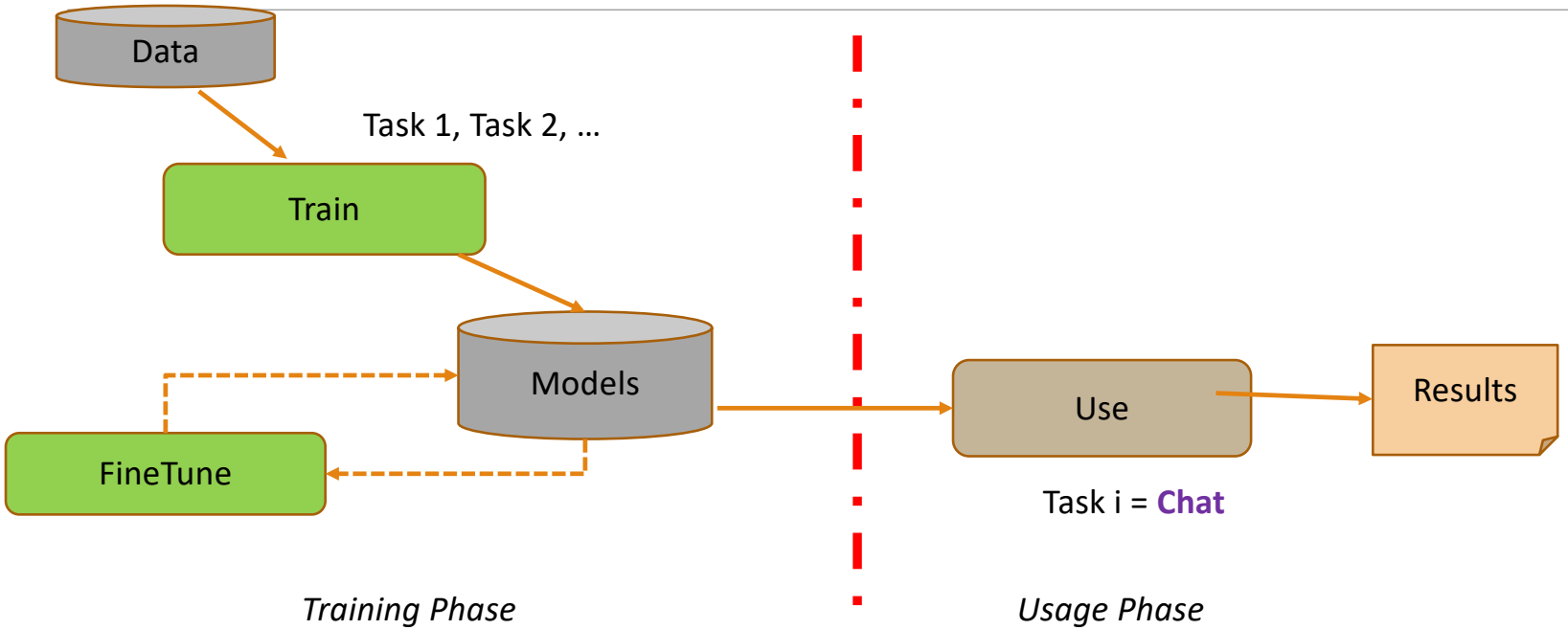
- *Prepare data – two datasets shown*
- *Try generator:*
  - *https://github.com/biplav-s/course-nl/blob/master/l7-language/code/TryMarkovifyLangModel.ipynb*

Large LMs (LLMs)

Large Language Models (LLMs) Basics



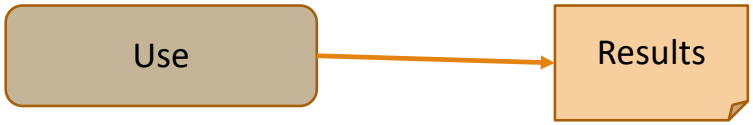
ChatGPT: Large Language Models (LLMs) based Chatbot



Another “Turning Point” Moment In Technology

Raised interest about Chatbots among public

- Excitement about new use-cases
- Concerns about social impact – cheating, jobs, misinformation
- Renewed calls for regulations



Task i = **Generally speaking: content generation – text, image, video, audio, …**

Usage Phase

BERT - Bidirectional Encoder Representations from Transformers

Learns with two tasks

- Predicting missing words in sentences
  - mask out 15% of the words in the input, predict the masked words.
- Given two sentences A and B, is B the actual next sentence that comes after A, or just a random sentence from the corpus?

(12-layer to 24-layer Transformer) on (Wikipedia + BookCorpus)

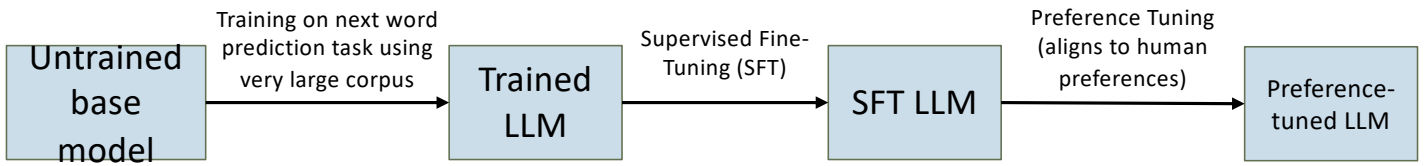
Input: the man went to the [MASK1] . he bought a [MASK2] of milk.
Labels: [MASK1] = store; [MASK2] = gallon

Sentence A: the man went to the store .
Sentence B: he bought a gallon of milk .
Label: IsNextSentence

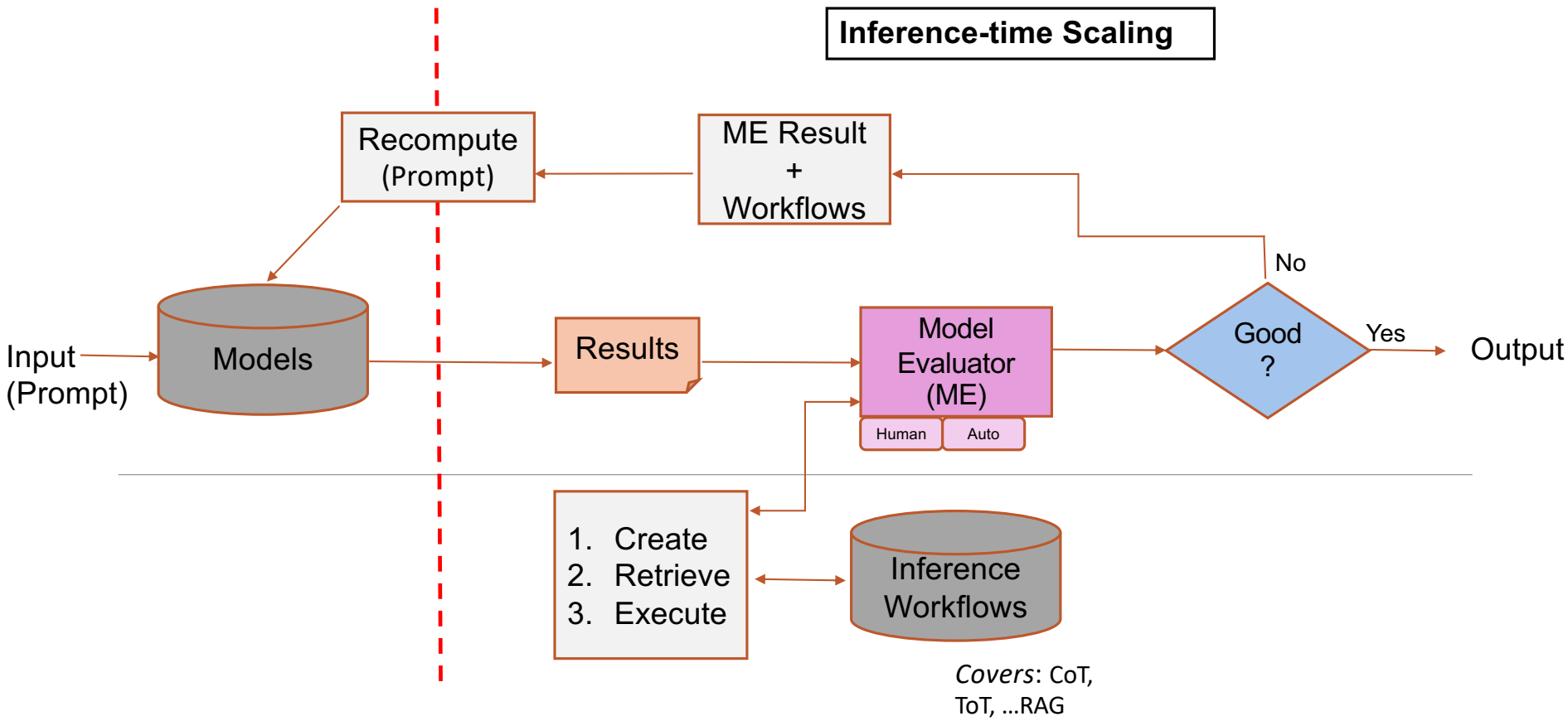
Sentence A: the man went to the store .
Sentence B: penguins are flightless .
Label: NotNextSentence

Credit and details: https://github.com/google-research/bert

LLM (Fine-tuning) Training Procedure



Inference Time with LLMs



LLM/ FM Tools

Using LLMs/FMs

- Choose a LLM/FM
  - Open-source: Llama, Mistral, DeepSeek, Bloom, …
  - Closed-source: GPT, Gemini, Claude, …
- API interface
  - Huggingface
  - Ollama
- Use chat interface
  - POE
  - ChatGPT
  - DeepSeek
  - …

Class 16: Project Update #2

Extra Task for Update #2

- Give project input to a LLM
- Collect its result
- Discuss
  - Is the result a good baseline?
  - Is your solution beating the baseline?
  - Where do you go from here?

Project Discussion

Project Milestones and Deliverables

- See course website for schedule and milestones: https://sites.google.com/site/biplavsrivastava/teaching/ai-csce-581-spring-2026-trusted-ai
  - Sprints: 1 (Crawl) : Feb 5-24, 2(Walk) : Feb 24-Mar31 and 3 (Run): Apr 2-21
  - Milestones – in-class presentations
    - Project discussion 1 – Feb 5
    - **Project discussion 2 – March 5**
    - Project presentations 3 (final) – April 21 and 23
      - Report due by April 23, 2026

Penalty

- Timelines of presentations [-10 points each miss; -30 total]
- Timelines of reports [-10 points]
- Code missing [-10 points]
- Testcase missing [-10 points]

Rubric for Evaluation of Course Project

Project

- Project plan along framework introduced (7 points)
- Challenging nature of project
- Actual achievement
- Report
- Sharing of code

Presentation

- Motivation
- Coverage of related work
- Results and significance
- Handling of questions

Penalty

- Timelines of presentations [-10 points each miss; -30 total]
- Timelines of reports [-10 points]
- Code missing [-10 points]
- Testcase missing [-10 points]

Course Project

- **Framework**
  1. (Problem) Think of a problem whose solution may benefit people (e.g., health, water, air, traffic, safety)
  2. (User) Consider how the primary user (e.g., patient, traveler) may be solving the problem today
  3. (AI Method) Think of what the solution will do to help the primary user
     1. Solution => ML task (e.g. classification), recommendation, text summarization, …
     2. Use a foundation model (e.g., LLM-based) solution as the baseline
  4. (Data) Explore the data for a solution to work
  5. (Reliability: Testing) Think of the evaluation metric we should employ to establish that the solution will works? (e.g., 20% reduction in patient deaths)
  6. (Holding Human Values) Discuss if there are fairness/bias, privacy issues?
  7. (Human-AI) Finally, elaborate how you will explain the primary user that your solution is trustable to be used by them

Project Discussion: What to Focus on ?

- Problem: you should care about it
- Data: should be available
- Method: you need to be comfortable with it. Have at least two – one serves as baseline
- Trust issue
  - Due to Users
    - Diverse demographics
    - Diverse abilities
    - Multiple human languages
  - Or other impacts
- What one does to mitigate trust issue

Project Discussion

1. Create a private Github repository called "CSCE581-Spring2025-<studentname>-Repo". Share with Instructor (biplav-s)
2. Create a folder called "Project". Inside, create a text file called "ProjectPlan.md" (or "ProjectPlan.txt") and have details by the next class (Jan 30, 2026)

1. Title:
2. Key idea: (2-3 lines)
3. Who will care when done:
4. Data need:
5. Methods:
6. Evaluation:
7. Users:
8. Trust issue:

Concluding Section

Week 8 (L15 and 16): Concluding Comments

- We looked at
  - Revised explanation methods
  - Did an overview of LLM/ FM basics and tools
  - Reviewed projects, especially in the context of a LLM/FM

About Next Week – Lectures 17, 18

Lectures 17, 18:

- Invited talk
- Text processing

13	Feb 24 (Tu)	AI - Supervised ML: Explanation Tools	W7 Project – Sprint 1 - end;
14	Feb 26 (Th)	AI Trust and Decision Making (Planning) – Bharath Muppasani	Project – Sprint 2 - start;
15	Mar 3 (Tu)	Large Language Models (LLMs), Machine Learning – Trust Issues (Explainability)	W8
16	Mar 5 (Th)	Student presentations - project	Semester – Midpoint Project Checkpoint
	Mar 10 (Tu)		W9 Spring break – No class
	Mar 12 (Th)		Spring break – No class
17	Mar 17 (Tu)	**Invited Guest** –	W10 Quiz 2 - start
18	Mar 19 (Th)	AI - Unstructured (Text): Processing and Representation	
19	Mar 24 (Tu)	AI - Unstructured (Text): Representation, Common NLP Tasks, Large Language Models (LLMs)	[Week 11] Quiz 2-end
20	Mar 26 (Th)	Natural Languages/ Language Models and their Impact on AI	
21	Mar 31 (Tu)	AI - Unstructured (Text): Analysis – Supervised ML – Trust Issues	[Week 12]Project – Sprint 2 - end